13)

$$C = \begin{bmatrix} 0 & 1 \\ 3 & 2 \\ -2 & 0 \end{bmatrix} \quad B = \begin{bmatrix} 1 & -3 & 0 \\ -3 & 1 & 0 \\ 0 & 0 & -2 \end{bmatrix} \quad C^T = \begin{bmatrix} 0 & 3 & -2 \\ 1 & 2 & 0 \end{bmatrix} \quad B^T = B$$

$$CC^T = \begin{bmatrix} 0 & 1 \\ 3 & 2 \\ -2 & 0 \end{bmatrix} \begin{bmatrix} 0 & 3 & -2 \\ 1 & 2 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} 0(0)+1(1) & 0(3)+1(2) & 0(-2)+1(0) \\ 3(0)+2(1) & 3(3)+2(2) & 3(-2)+2(0) \\ -2(0)+0(1) & -2(3)+0(2) & -2(-2)+0(0) \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 2 & 0 \\ 2 & 13 & -6 \\ 0 & -6 & 4 \end{bmatrix}$$

$$BC = \begin{bmatrix} 1 & -3 & 0 \\ -3 & 1 & 0 \\ 0 & 0 & -2 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 3 & 2 \\ -2 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} 1(0)+-3(3)+0(-2) & 1(1)+-3(2)+0(0) \\ -3(0)+1(3)+0(-2) & -3(1)+1(2)+0(0) \\ 0(0)+0(3)+-2(-2) & 0(1)+0(2)+0(0) \end{bmatrix}$$

$$= \begin{bmatrix} -9 & -5 \\ 3 & -1 \\ 4 & 0 \end{bmatrix}$$

(c)

$$CB = \begin{bmatrix} 0 & 1 \\ 3 & 2 \\ -2 & 0 \end{bmatrix}\begin{bmatrix} 1 & -3 & 0 \\ -3 & 1 & 0 \\ 0 & 0 & -2 \end{bmatrix}$$

$3\times 2 \qquad 3\times 3$

$$= \boxed{\text{UNDEFINED}}$$

Because matrices of size 3×2 and 3×3 can be multiplied in this order

$$C^TB = \begin{bmatrix} 0 & 3 & -2 \\ 1 & 2 & 0 \end{bmatrix}\begin{bmatrix} 1 & -3 & 0 \\ -3 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

$$= \begin{bmatrix} 0(1)+3(-3)+-2(0) & 0(-3)+3(1)+-2(0) & 0(0)+3(0)+2(2) \\ 1(1)+2(-3)+0(0) & 1(-3)+2(1)+0(0) & 1(0)+2(0)+0(2) \end{bmatrix}$$

$$\boxed{= \begin{bmatrix} -9 & 5 & 4 \\ -5 & -1 & 0 \end{bmatrix}}$$

1)

$$4x - 6y = -11$$
$$-3x + 8y = 10$$

$$\left[\begin{array}{cc|c} 4 & -6 & -11 \\ -3 & 8 & 10 \end{array}\right] \Rightarrow$$

$$\left[\begin{array}{cc|c} 1 & 2 & -1 \\ -3 & 8 & 10 \end{array}\right] \Rightarrow$$

$$\left[\begin{array}{cc|c} 1 & 2 & -1 \\ 0 & 14 & 7 \end{array}\right] \Rightarrow$$

$$\left[\begin{array}{cc|c} 1 & 2 & -1 \\ 0 & 2 & 1 \end{array}\right] \Rightarrow$$

$$\left[\begin{array}{cc|c} 1 & 2 & -1 \\ 0 & 1 & 1/2 \end{array}\right] \Rightarrow$$

$$\left[\begin{array}{cc|c} 1 & 0 & -2 \\ 0 & 1 & 1/2 \end{array}\right] \Rightarrow \boxed{x = -2,\ y = 1/2}$$

1)

$$\begin{bmatrix} 4 & -2 & 6 \\ -2 & 1 & -3 \end{bmatrix} \Rightarrow$$

$$\begin{bmatrix} 4 & -2 & 6 \\ -4 & 2 & -6 \end{bmatrix} \Rightarrow$$

$$\begin{bmatrix} 4 & -2 & 6 \\ 0 & 0 & 0 \end{bmatrix} \Rightarrow$$

Rank = 1

2)

$$3x - 5y = 15.5$$
$$6x + 16y = 5.0$$

$$D = \begin{vmatrix} 3 & -5 \\ 6 & 16 \end{vmatrix} \qquad D_x = \begin{vmatrix} 15.5 & -5 \\ 5 & 16 \end{vmatrix} \qquad D_y = \begin{vmatrix} 3 & 15.5 \\ 6 & 5 \end{vmatrix}$$

$= 3(16) - -5(6)$ $\quad D_x = 15.5(16) - -5(5)$ $\quad D_y = 3(5) - 15.5(6)$

$= 78$ $\quad = 273$ $\quad = -78$

$$X = \frac{D_x}{D} = \frac{273}{78} = \boxed{3.5} \qquad y = \frac{D_y}{D} = \frac{-78}{78} = \boxed{-1}$$

5|

$$A=\begin{bmatrix}1&0&0\\2&1&0\\5&4&1\end{bmatrix} \Rightarrow$$

$$\left[\begin{array}{ccc|ccc}1&0&0&1&0&0\\2&1&0&0&1&0\\5&4&1&0&0&1\end{array}\right] \Rightarrow$$

$$\left[\begin{array}{ccc|ccc}1&0&0&1&0&0\\0&1&0&-2&1&0\\0&4&1&-5&0&1\end{array}\right] \Rightarrow$$

$$\left[\begin{array}{ccc|ccc}1&0&0&1&0&0\\0&1&0&-2&1&0\\0&0&1&-5&-4&1\end{array}\right] \Rightarrow$$

$$\boxed{A^{-1}=\begin{bmatrix}1&0&0\\-2&1&0\\-5&-4&1\end{bmatrix}}$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 5 & 4 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ -5 & -4 & 1 \end{bmatrix} = \begin{bmatrix} 1(1)+0(-2)+0(-5) & 1(0)+0(1)+0(-4) & 1(0)+0(0)+0(1) \\ 2(1)+1(-2)+0(-5) & 2(0)+1(1)+0(-4) & 2(0)+1(0)+0(1) \\ 5(1)+4(-2)+1(1) & 5(0)+4(1)+1(-4) & 5(0)+4(0)+1(1) \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

6)

$$\begin{bmatrix} 0.2 & -0.1 & 0.4 \\ 0 & 1.1 & -0.3 \\ 0.1 & 0 & -2.1 \end{bmatrix} \Rightarrow$$

$$\begin{bmatrix} 2 & -1 & 4 \\ 0 & 11 & -3 \\ 1 & 0 & -21 \end{bmatrix} \Rightarrow$$

$$\begin{bmatrix} 1 & 0 & -21 \\ 0 & 11 & -3 \\ 2 & -1 & 4 \end{bmatrix} \Rightarrow$$

$$\begin{bmatrix} 1 & 0 & -21 \\ 0 & 11 & -3 \\ 0 & -1 & 46 \end{bmatrix} \Rightarrow$$

$$\boxed{\text{Rank} = 3}$$

$$\begin{bmatrix} 1 & 0 & -21 \\ 0 & 1 & -46 \\ 0 & 11 & -3 \end{bmatrix} \Rightarrow$$

$$\begin{bmatrix} 1 & 0 & -21 \\ 0 & 1 & -46 \\ 0 & 0 & 503 \end{bmatrix} \Rightarrow \text{Divide Row 3 by 503 and eliminate the third column} \Rightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$